



Comparison of machine learning models for predicting 30-day readmission rates for patients with diabetes

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Background: Affecting over 25.5 million individuals in the United States, diabetes is linked to increased hospital readmission rates, almost double that of patients without diabetes. Previous research suggested the predictive superiority of deep learning (DL) models, particularly long short-term memory (LSTM) models, compared to traditional machine learning (ML) models, in forecasting diabetes-related hospital readmissions. The objective of this study was to compare one DL and 10 ML methods for predicting 30-day readmission rates in patients with diabetes.

Methods: The dataset was obtained from “Diabetes 130-US Hospitals for Years 1999–2008” in the University of California Irvine (UCI) Machine Learning Repository and consisted of 101,766 unique encounters from 130 hospitals and integrated delivery networks across the United States. Encounters were included if they were an inpatient encounter (i.e., hospital admission) during which diabetes was entered in the system as a diagnosis. Predictors of readmission included demographic variables (age, race, gender), admission and discharge type, number of procedures and medications, diagnosis, and hemoglobin A1c (HbA1c) testing, a measure of the last 3 months of glycemic control. The outcome was the rate of readmission within 30 days. This study built upon traditional ML algorithms by incorporating the Grey Wolf Optimizer (GWO), a swarm intelligence-based optimization algorithm, for feature selection. Data preprocessing involved handling missing values, encoding categorical variables, and converting the target variable into a binary classification task. Feature engineering included creating service utilization variables, age group midpoints, and grouping admission types. Categorical features were one-hot encoded, and the dataset underwent standard scaling. Feature selection was conducted using the GWO algorithm, and class imbalance was addressed with synthetic minority over-sampling technique (SMOTE) during group k-fold cross-validation. Eleven ML algorithms, including random forest (RF), extreme gradient boosting (XGBoost), decision tree, support vector machine (SVM), and LSTM were employed for model training and evaluation.

Results: RF emerged as the most favorable model, consistently outperforming others in F1 score, accuracy, precision score, and recall score. The RF model achieved the highest F1 score (0.83) and accuracy (0.88), indicating its superior predictive ability, especially in balancing precision and recall. XGBoost showed comparable results, achieving the second highest F1 score (0.84) and accuracy (0.88).

Conclusions: The RF and XGBoost models with GWO outcompeted previous DL predictive modeling in diabetes hospital readmission scenarios. However, ML with GWO adoption should be carefully considered based on specific hospital needs, patient populations, and available resources, acknowledging ongoing advancements in predictive analytics for healthcare. Other advanced algorithms, including DL, still play a pivotal role. ML with GWO represents progress in predictive analytics for hospital readmission scenarios.

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Introduction

Background

Affecting over 25.5 million people in the United States, with the prevalence increasing over time (1), diabetes is associated with an increased hospital readmission rate of 14.4–22.7%, almost twice that of patients without diabetes (8.5–13.5%) (2,3). Readmission for this study was defined as the all-cause re-hospitalization of a patient to any of the 130 hospitals and integrated delivery networks within the system, both planned and unplanned (4). Higher hospital

readmission rates indicate an increased disease burden for patients with diabetes and have important medical, social, quality of life, and financial implications (5). In one study, poor glycemic control was found to be the most significant predictor for hospitalization among people with diabetes (6), while another study found male sex, longer duration of prior hospitalization, number of previous hospitalizations, number and severity of comorbidities, and lower socioeconomic and/or educational status to be risk factors for readmission (2). Another study also found that an increased number of prior hospital admissions was the most important predictor of readmission, emphasizing the importance of longitudinal data (4).

Several studies have attempted to predict diabetes-related hospital readmissions. In a recent study by Hai *et al.*, the researchers employed deep learning (DL) long short-term memory (LSTM) models to predict unplanned, all-cause, 30-day readmissions among 36,641 diabetes patients (4). They compared these DL models with traditional models, and the LSTM model demonstrated significantly higher performance, with an area under the receiver operating characteristic (ROC) curve (AUROC) of 0.79 compared to random forest (RF) at 0.72 ($P < 0.0001$). The study highlights the superiority of DL models in predicting readmission risk (4). The LSTM model's performance improved with an increasing number of prior encounters, showcasing its potential for robust predictions.

DL models leverage deep neural architectures to automatically learn hierarchical representations from data, whereas traditional machine learning (ML) models often rely on explicit feature engineering and may not handle complex patterns as effectively in certain scenarios (7). The choice between them depends on the nature of the data, the size of the dataset, and the specific requirements of the task at hand.

Shang *et al.*'s study focused on predicting 30-day hospital readmission risk in patients with diabetes using traditional ML classifiers (8). The dataset included over 100,000 records, and the researchers applied various classifiers such as RF, Naïve Bayes, and decision tree ensemble. The RF

Highlight box

Key findings

- To predict diabetes-related hospital readmissions, this study compared traditional machine learning (ML) algorithms using Grey Wolf Optimizer (GWO). The random forest (RF) model consistently outperformed others, with the extreme gradient boosting (XGBoost) classifier also showcasing competitive performance. These findings emphasize the relevance of advanced modeling techniques, feature selection, and addressing class imbalance for accurate predictions.

What is known and what is new?

- Previous research explored models such as deep learning long short-term memory and traditional classifiers for diabetes readmissions.
- This study introduces the GWO algorithm and comprehensively compares 10 ML algorithms. It contributes to the existing literature by demonstrating the superiority of the RF model and the XGBoost classifier's F1 score, revealing that deep learning is not the only predictive modeling useful for diabetes readmissions.

What is the implication, and what should change now?

- The study's implications for healthcare providers and policymakers are significant. The success of the RF model and the competitive performance of the XGBoost classifier when combined with GWO suggest that hospitals should consider adopting advanced ML algorithms to enhance risk assessments and improve patient outcomes. The findings underscore the need for an evolving approach to predictive analytics, encouraging healthcare institutions to incorporate advanced algorithms for more effective and personalized patient care.

Table 1 Strengths and weaknesses of ML models

ML model	Description
SVM linear kernel	SVMs with a linear kernel are effective in high-dimensional spaces and resistant to overfitting. However, they may struggle with noisy data and large datasets
SVM with RBF kernel	SVM with an RBF kernel can handle non-linear decision boundaries but requires careful hyperparameter tuning and can be computationally demanding. Several kernel functions are available, designed for different data types
XGBoost classifier	XGBoost classifier excels in predictive accuracy and performs well on structured data. It is robust to outliers and can handle missing data effectively. Nevertheless, it may require extensive hyperparameter tuning and can be computationally expensive
Decision tree	Decision trees offer simplicity and interpretability but are susceptible to overfitting and can be unstable due to small changes in the data
RF	RFs address the overfitting issue of individual decision trees and are robust across various data types. They provide insights into feature importance but might be computationally expensive and struggle with highly imbalanced datasets
Logistic regression	Logistic regression is a powerful and interpretable method for binary classification, suitable for modeling the probability of an event and efficient for linearly separable problems; however, its assumptions of linearity and independence might be restrictive in capturing complex relationships in the data where a linear decision boundary does not suffice
KNN	KNN is simple and intuitive, but it becomes computationally expensive during prediction, particularly with large datasets because it stores the entire training dataset without undergoing a separate training stage. It is also sensitive to irrelevant or redundant features
Naïve Bayes	Naïve Bayes, known for its simplicity and efficiency, assumes independence between features, which might limit its performance on data with complex dependencies
AdaBoost	AdaBoost combines weak learners to form a strong learner, offering resilience against overfitting. However, it can be sensitive to noisy data and outliers, and training time may be relatively high
MLP	The MLP excels at capturing complex relationships in data and is adaptable to various types of problems. However, its disadvantages include vulnerability to overfitting, dependence on large datasets, and the need for careful tuning of hyperparameters. MLP is known for its capability to model complex non-linear relationships, making it effective in tasks like image and speech recognition. However, it is prone to overfitting and demands careful tuning of hyperparameters. Additionally, it is sensitive to feature scaling

ML, machine learning; SVM, support vector machine; RBF, radial basis function; XGBoost, extreme gradient boosting; RF, random forest; KNN, k-nearest neighbor; AdaBoost, adaptive boosting; MLP, multilayer perceptron.

model exhibited the best performance, achieving a higher AUROC compared to other algorithms (8).

Both studies contribute valuable insights into the factors influencing hospital readmission in diabetes patients. Hai *et al.*'s research emphasized the superiority of DL models (4), while Shang *et al.*'s study demonstrated that traditional ML classifiers, particularly the RF model, are still viable and effective for predicting readmission risk (8). These findings collectively underscore the importance of advanced modeling techniques and comprehensive feature selection in enhancing the accuracy of readmission risk prediction for patients with diabetes, thereby facilitating more targeted and effective interventions to improve patient outcomes.

Rationale and knowledge gap

Using hospital admission data from 130 hospitals and integrated delivery networks throughout the United States (9), we examined differences in hospital readmissions for diabetes-related conditions. Our study expands on past prognostic studies examining 30-day readmission rates. Fifty patient variables were available for analysis [see *Table 1* of Strack *et al.* (9)]. According to prior literature, readmissions observed over a longer period of time are primarily related to progression of chronic disease, such as diabetes, and thus serve as a measure of the quality of outpatient care, whereas readmissions occurring shortly after a hospital stay are

related to quality-of-care issues during the initial admission (10,11).

While several studies have endeavored to forecast diabetes-related hospital readmissions, a notable gap exists in comparing the predictive capabilities of traditional ML models with emerging DL techniques. Recent research by Hai *et al.* (4) employed DL LSTM models to forecast unplanned, all-cause, 30-day readmissions among a sizable cohort of diabetes patients. Their findings highlighted the superior performance of DL models, particularly the LSTM model, compared to conventional approaches such as RF. However, the potential of traditional ML classifiers, particularly in predicting readmission risk, warrants further exploration.

Additionally, the incorporation of innovative optimization algorithms, such as the Grey Wolf Optimizer (GWO), presents an opportunity to enhance the predictive accuracy of traditional ML models in this domain. A tradeoff exists between including all features in a model, which results in high dimensionality and more computationally expensive models, and involving a domain expert to select a subset of variables, which can be costly and less feasible (4). Automatic feature selection can strike a balance between the two goals of a parsimonious and a complete set of variables. GWO, a metaheuristic, nature-inspired optimization algorithm, has been used for feature selection in other contexts with promising results (12). The importance of a robust feature selection method in the preprocessing stage was emphasized by Cui *et al.*, who found that a novel feature selection technique in combination with SVM produced their best model (13).

There has been broad interest in readmission, but relatively little research has focused specifically on readmission of patients with diabetes (2). This contrasts with the great potential that reducing diabetes-related readmissions has to decrease healthcare expenditures: Agency for Healthcare Research and Quality Nationwide Inpatient Sample data from 2012 indicated that a modest 5% reduction in the 30-day readmission rate would result in 82,754 fewer admissions per year and an annual cost savings of \$1.2 billion (2,8). As it is already clear that diabetic patients have an increased 30-day readmission rate of 14.4–22.7% *vs.* 8.5–13.5% for the overall readmission rate of hospitalized patients, this study focuses on predicting readmission for diabetes patients, rather than comparing patients with and without diabetes.

Objective

The primary objective of this study was to evaluate and compare the predictive performance of DL and traditional ML algorithms in forecasting diabetes-related hospital readmissions using a primary measure of F1 score. In developing and validating 10 ML models, we aimed to assess the efficacy of DL LSTM models against a selection of traditional ML classifiers, including support vector machine (SVM), extreme gradient boosting (XGBoost), decision tree, RF, logistic regression, k-nearest neighbor (KNN), Naïve Bayes, adaptive boosting (AdaBoost), multilayer perceptron (MLP), and SVM with radial basis function (RBF) kernel. Furthermore, we sought to integrate the GWO optimization algorithm into traditional ML frameworks to ascertain its impact on predictive accuracy. By elucidating the strengths and limitations of both DL and traditional ML approaches, this study aimed to provide insights into optimizing readmission risk prediction models for patients with diabetes, thereby facilitating more targeted interventions and improved patient outcomes.

Non-standardized inpatient and outpatient diabetes care can lead to poor glycemic control, which increases management and care costs and can contribute to patient readmissions. In the United States in 2022, the cost of care for people diagnosed with diabetes accounted for one in four healthcare dollars, with patients with diabetes spending \$19,736 per year on medical expenses, about 2.6 times higher than patients without diabetes (1). Diabetes-related problems, such as microvascular complications (neuropathy, nephropathy, retinopathy) and macrovascular complications (such as cardiovascular disease and stroke) may also increase patients' morbidity and death. The study seeks to address these concerns and enhance patient care results. We present this article in accordance with the TRIPOD reporting checklist (available at <https://jmai.amegroups.com/article/view/10.21037/jmai-24-70/rc>).

Methods

Description of dataset and variables

Our study focused primarily on predictive modeling for hospital readmissions, utilizing a dataset obtained from the University of California Irvine (UCI) Machine Learning Repository titled “Diabetes 130-US hospitals for years 1999–2008” (14). The dataset includes a decade

of clinical treatment data [1999–2008] from the Health Facts Database, with a specific focus on patients with diabetes, from 130 United States hospitals and integrated delivery networks in the Midwest (18 hospitals), Northwest (58 hospitals), South (28 hospitals), and West (16 hospitals). The original dataset comprises 74,036,643 unique encounters for 17,880,231 unique patients and 2,889,571 providers. The encounter satisfied the inclusion criteria if it was an inpatient encounter (i.e., hospital admission), any kind of diabetes was entered in the system as a diagnosis, the length of stay was between 1 and 14 days, and laboratory tests and medications were administered. Each entry in the dataset from the UCI ML repository thus included hospital information test results, prescriptions, and up to 14-day hospitalization records for the patients.

Our main objective was to predict the rates of early readmissions, defined as within 30 days after discharge. 30 days is a criterion usually used by funding agencies (9). Inadequate diabetes care in hospitals can lead to worse glycemic control, increased management costs due to more frequent patient readmissions and an increased risk of morbidity and mortality among these patients, and hospitals have been fined for having greater than expected 30-day readmission rates (2,8).

Factors included in modeling were demographics (age, race, gender), admission and discharge type, number of procedures and medications, diagnosis, and hemoglobin A1c (HbA1c) testing, a measure of the preceding 3 months of glycemic control. These were measured and stored in the Health Facts electronic health records system at the time of the encounter and included encounter data (emergency, inpatient), provider specialty, demographics (age, sex, and race), diagnoses and in-hospital procedures documented by the International Classification of Diseases, 9th Revision, Clinical Modification (ICD-9-CM) codes, laboratory data, pharmacy data, in-hospital mortality, and hospital characteristics (9). The ‘diabetes medication’ variable was a binary variable indicating whether any of the 24 generic medications (metformin, repaglinide, etc.) was prescribed. Each of the medications was associated with value ‘up’ if the dosage was increased during the encounter, ‘down’ if the dosage was decreased, ‘steady’ if the dosage did not change, and ‘no’ if the drug was not prescribed (9). All records were deidentified. *Table 1* of Strack *et al.* contains a detailed description of all these inputs (9).

ML models and preprocessing

We selected 10 ML algorithms, including SVM with linear kernel, SVM with RBF kernel, XGBoost classifier, decision tree, RF, logistic regression, KNN, Naïve Bayes, AdaBoost, and MLP. We also selected one DL algorithm, which was LSTM.

The above-selected ML models vary in their strengths and weaknesses. *Table 1* contains a summary of each ML model’s strengths and weaknesses (15–20). XGBoost classifier excels in predictive accuracy and performs well on structured data (21). It is robust to outliers and can handle missing data effectively. Nevertheless, it may require extensive hyperparameter tuning and can be computationally expensive.

Decision trees offer simplicity and interpretability but are susceptible to overfitting and can be unstable due to small changes in the data (22). RFs address the overfitting issue of individual decision trees and are robust across various data types (23). They provide insights into feature importance but might be computationally expensive and struggle with highly imbalanced datasets.

Logistic regression is a powerful and interpretable method for binary classification, suitable for modeling the probability of an event and efficient for linearly separable problems (24); however, its assumptions of linearity and independence might be restrictive in capturing complex relationships in the data where a linear decision boundary does not suffice.

For this research, we employed the zoofs library version 0.1.26 to implement the GWO algorithm for feature selection (25). The library and its dependencies were installed using the pip package manager.

We conducted several data preprocessing steps, including handling missing values, encoding categorical variables, and converting the target variable into a binary classification task. Additionally, we performed various transformations and mappings on the dataset, such as re-encoding admission type, discharge type, and admission source into fewer categories. We converted age intervals into mid-values and eliminated duplicate patient records based on the ‘patient_nbr’ column. Feature engineering techniques were employed to process diagnosis columns, resulting in the creation of two sets of diagnostic levels. Each diagnosis level was then mapped to a numerical representation based

on specific conditions. Further modifications to the dataset involved calculating the total number of medications used by each patient and removing some granular diagnosis columns. Notably, we created a service utilization variable based on the sum of outpatient visits, emergency visits, and inpatient visits. Additionally, we grouped admission type and admission source into similar categories. Moreover, we performed data preprocessing tasks such as removing columns with high missing values and only one unique value, as well as grouping the categories of “30-day+” and “non-readmitted” into a single category labeled “non-readmitted within 30 days”.

Categorical features were one-hot encoded, and the dataset was divided into numerical and categorical components. Subsequently, standard scaling was applied to the numerical features. Feature selection was carried out using the GWO algorithm from the *zoofs* library. The selected features were utilized for model training and evaluation. To avoid biasing the results, we employed group k-fold cross-validation to group encounters for the same patient within the same fold. Synthetic minority over-sampling technique (SMOTE) was applied on the training set of each fold to address class imbalance, resulting in a balanced representation of readmitted and non-readmitted cases (26).

A sequential LSTM model was specified with two LSTM layers with 50 units followed by dropout layers with a dropout rate of 0.2. These were followed by a dense layer with a single unit and a sigmoid activation function. The model was compiled using binary cross entropy as the loss function and the Adam optimizer. The default hyperparameters were used for most traditional ML models. For the MLP classifier and logistic regression, a maximum of 1,000 iterations was specified. The XGBoost classifier was configured to use the log loss evaluation metric. The Keras classifier for the LSTM model was trained for 10 epochs with a batch size of 32.

The performance of various ML models, including MLP, XGBoost classifier, decision tree, RF, logistic regression, SVM (linear and RBF kernel), KNN, Naïve Bayes, and AdaBoost, was evaluated. Performance metrics such as F1 score, accuracy, precision score, recall score, false negatives, and AUROC were computed for each model across 5 folds. The average metrics were summarized in a table, providing insights into the comparative performance of the models in predicting hospital readmission.

Statistical analysis

Figure 1 illustrates a comprehensive overview of a decade-long dataset spanning from 1999 to 2008, consisting of electronic health records gathered from 130 hospitals across the United States. This extensive dataset comprises 117 distinct features and encompasses a staggering 74,036,643 unique patient encounters.

The progression of data refinement and processing stages, started with the establishment of inclusion criteria focused on features associated with diabetes. The dataset was preprocessed to remove duplicates and handle missing values, culminating in the extraction of 55 pertinent features from 101,776 encounters, some of which represent readmissions.

Following data preprocessing, feature engineering further refined the dataset by dropping columns with excessive missing values, only one unique value, and creating a new variable called service utilization. Utilizing the Greywolf optimizer for feature selection from the *zoofs* package, the dataset was streamlined to 43 features, yet retained the same number of encounters. *Figure 1* also illustrates the stratification of the data into folds for model training and testing purposes, with the dataset divided into 5 folds. Notably, each fold comprises varying numbers of training and testing instances.

We built upon traditional ML algorithms to see if we could achieve comparable or better results than the DL algorithms in predicting diabetes readmission by incorporating the GWO, a swarm intelligence-based optimization algorithm, that was developed by Mirjalili *et al.* in 2014 (25). It is inspired by the social hierarchy and hunting behavior of grey wolves in nature. The algorithm mimics the leadership progression of wolves, where the alpha wolf is responsible for making decisions, and the beta and delta wolves assist the alpha in fundamental administration. The remaining wolves are considered omega wolves. In GWO, the fitness solution is known as the alpha, and the second, third, and fourth most excellent solutions are named beta, delta, and omega, respectively. The algorithm has been successfully applied in various optimization problems, including feature selection.

Ultimately, *Figure 1* shows that the results were averaged across 5 folds to derive a more accurate, less biased estimate of a model's performance, and aid in model selection. These comprehensive evaluations of model performance included

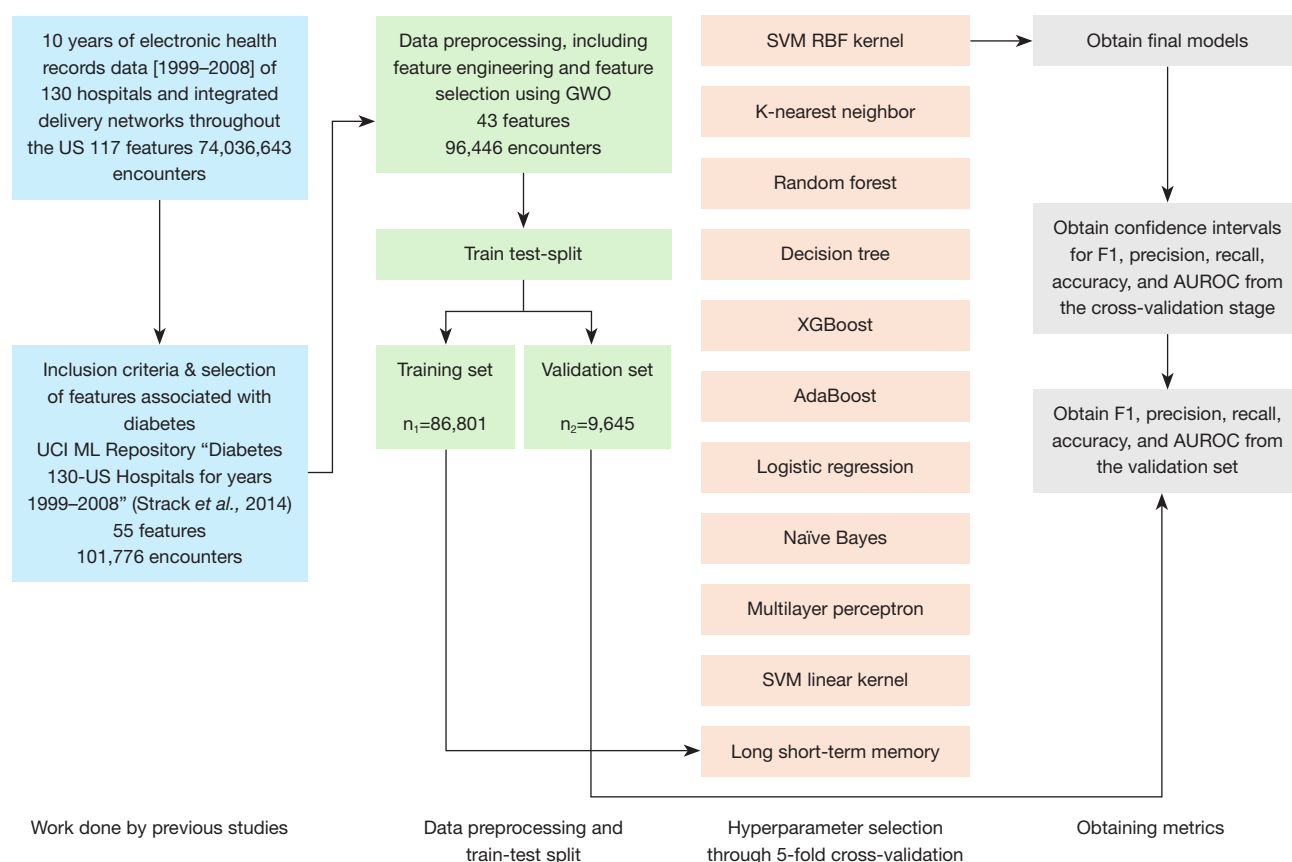


Figure 1 Patient inclusion criteria and distribution into train and test fold datasets. US, United States; UCI, University of California Irvine; ML, machine learning; GWO, Grey Wolf Optimizer; SVM, support vector machine; RBF, radial basis function; XGBoost, extreme gradient boosting; AUROC, area under the receiver operating characteristic curve.

F1 score, accuracy, precision, recall, and AUROC for each ML algorithm evaluated. The statistical analysis was conducted in Python programming language.

Results

Information about patient demographics, hospital admission details, lab testing, and medication use among the 101,766 patients included in this study are shown in Table 2, presented overall and also broken down by early readmission (within 30 days) and later readmission (more than 30 days) or not readmitted. Notably, most patients were above 60 years of age (>65%), Caucasian (>75%), female (53.8%), and admitted through the emergency department (53%). On average, patients spent just over 4 days (4.4 days) in the hospital before being discharged home. Information on hospital department, discharge disposition, mean number of medications taken, lab tests,

procedures done, and HbA1c are also included. In total, 11,357 of the 101,766 patients were readmitted within 30 days of discharge.

The results of applying various ML models to predict diabetes readmissions are presented in Table 3, a comprehensive overview of the ML models' performance across different evaluation metrics. Each row corresponds to a specific model, and the columns represent key metrics of F1 score, accuracy, precision score, recall score, and AUROC. The overall results in Table 3 were obtained by averaging the score across the 5 folds for each of the metrics F1 score, accuracy, precision, recall, and AUROC. Regression coefficients were not available, but by following the preprocessing steps, the ML models may be fully reproducible. The small width of the confidence intervals (CIs) underscores the reliability of the estimates.

The F1 score, a metric that balances precision and recall and the primary outcome measure of the study, ranged

Table 2 Patient demographics, admission/discharge and lab test characteristics

Variables	Overall, n (%)	Readmitted in <30 days, n (%)	Readmitted in >30 days or not readmitted, n (%)
Total	101,766 (100.0)	11,357 (100.0)	90,409 (100.0)
Gender			
Female	54,708 (53.8)	6,152 (54.2)	48,556 (53.7)
Male	47,055 (46.2)	5,205 (45.8)	41,850 (46.3)
Age (years)			
0–9	161 (0.2)	3 (0.03)	158 (0.2)
10–19	691 (0.7)	40 (0.4)	651 (0.7)
20–29	1,657 (1.6)	236 (2.1)	1,421 (1.6)
30–39	3,775 (3.7)	424 (3.7)	3,351 (3.7)
40–49	9,685 (9.5)	1,027 (9.0)	8,658 (9.6)
50–59	17,256 (17.0)	1,668 (14.7)	15,588 (17.2)
60–69	22,483 (22.1)	2,502 (22.0)	19,981 (22.1)
70–79	26,068 (25.6)	3,069 (27.0)	22,999 (25.4)
80–89	17,197 (16.9)	2,078 (18.3)	15,119 (16.7)
90–99	2,793 (2.7)	310 (2.7)	2,483 (2.7)
Race			
Caucasian	76,099 (74.8)	8,592 (75.7)	67,507 (74.7)
African American	19,210 (18.9)	2,155 (19.0)	17,055 (18.9)
Asian	641 (0.6)	65 (0.6)	576 (0.6)
Hispanic	2,037 (2.0)	212 (1.9)	1,825 (2.0)
Other	1,506 (1.5)	145 (1.3)	1,361 (1.5)
Admission type			
Emergency	53,990 (53.1)	6,221 (54.8)	47,769 (52.8)
Urgent	18,480 (18.2)	2,066 (18.2)	16,414 (18.2)
Elective	18,869 (18.5)	1,961 (17.3)	16,908 (18.7)
Primary diagnosis			
Diseases of the circulatory system	30,437 (29.9)	3,485 (30.7)	26,952 (29.8)
Diseases of the respiratory system	14,423 (14.2)	1,403 (12.4)	13,020 (14.4)
Diseases of the digestive system	9,475 (9.3)	1,015 (8.9)	8,460 (9.4)
Endocrine, nutritional and metabolic diseases, and immunity disorders	8,757 (8.6)	1,137 (10.0)	7,620 (8.4)
Injury and poisoning	6,974 (6.9)	854 (7.5)	6,120 (6.8)
Diseases of the musculoskeletal system and connective tissue	4,957 (4.9)	471 (4.1)	4,486 (5.0)
Diseases of the genitourinary system	5,117 (5.0)	555 (4.9)	4,562 (5.0)
Neoplasms	3,433 (3.4)	346 (3.0)	3,087 (3.4)

Table 2 (continued)

Table 2 (continued)

Variables	Overall, n (%)	Readmitted in <30 days, n (%)	Readmitted in >30 days or not readmitted, n (%)
Time in hospital (days), mean	4.4	4.8	4.3
Discharge disposition			
Home	60,234 (59.2)	5,602 (49.3)	54,632 (60.4)
Skilled nursing facility	13,954 (13.7)	2,046 (18.0)	11,908 (13.2)
Home health service	12,902 (12.7)	1,638 (14.4)	11,264 (12.5)
Department			
Internal medicine	14,635 (14.38)	1,646 (14.49)	12,989 (14.37)
Family/general practice	7,440 (7.31)	883 (7.77)	6,557 (7.25)
Surgery-general	3,099 (3.05)	342 (3.01)	2,757 (3.05)
Number of procedures done, mean	1.3	1.3	1.3
Number of diagnoses, mean	7.4	7.7	7.4
Number of lab procedures, mean	43.1	44.2	42.7
HbA1c			
>8%	8,216 (8.1)	811 (7.1)	7,405 (8.2)
>7%	3,812 (3.7)	383 (3.4)	3,429 (3.8)
Normal (4.0–5.6%)	4,990 (4.9)	482 (4.2)	4,508 (5.0)
None	84,748 (83.3)	9,681 (85.2)	75,067 (83.0)
Serum glucose			
>300 mg/dL	1,264 (1.2)	181 (1.6)	1,083 (1.2)
>200 mg/dL	1,485 (1.5)	185 (1.6)	1,300 (1.4)
Normal (70–140 mg/dL)	2,597 (2.6)	295 (2.6)	2,302 (2.5)
None	96,420 (94.7)	10,696 (94.2)	85,724 (94.8)
Diabetes meds			
Yes	78,363 (77.0)	9,111 (80.2)	69,252 (76.6)
No	23,403 (23.0)	2,246 (19.8)	21,157 (23.4)
Medications, mean	16	16.9	15.9
HbA1c, hemoglobin A1c.			

from 0.03 for Naïve Bayes (95% CI: 0.03, 0.03) to 0.83 for RF (95% CI: 0.83, 0.84) and 0.84 for XGBoost (95% CI: 0.83, 0.84). This metric reflects the models' effectiveness in achieving a balance between minimizing false positives and false negatives. The highest F1 score was observed for the RF model, indicating its superior ability to provide accurate predictions while maintaining a good balance between precision and recall.

Accuracy, which measures overall correctness, mirrors

the trends in the F1 score. Again, RF achieved the highest accuracy (0.88, 95% CI: 0.88, 0.89), emphasizing its proficiency in making correct predictions across the dataset. Logistic regression and Naïve Bayes exhibited lower accuracy, indicating a higher rate of misclassifications in their predictions.

Precision score, representing the ability to correctly identify positive instances among predictions, ranged from 0.81 (MLP, decision tree, and KNN) to 0.83 for RF (95%

Table 3 Performance of the eleven ML models for predicting diabetes hospital readmissions with 95% CIs

ML model	F1 (95% CI)	Accuracy (95% CI)	Precision score (95% CI)	Recall score (95% CI)	AUROC (95% CI)
MLP	0.79 (0.78, 0.8)	0.77 (0.77, 0.78)	0.81 (0.81, 0.82)	0.77 (0.77, 0.78)	0.58 (0.57, 0.59)
XGBoost classifier	0.84 (0.83, 0.84)	0.88 (0.88, 0.89)	0.83 (0.83, 0.84)	0.88 (0.88, 0.89)	0.64 (0.64, 0.65)
Decision tree	0.80 (0.8, 0.8)	0.79 (0.79, 0.79)	0.81 (0.8, 0.81)	0.79 (0.79, 0.79)	0.52 (0.52, 0.53)
RF	0.83 (0.83, 0.84)	0.88 (0.88, 0.89)	0.83 (0.82, 0.83)	0.88 (0.88, 0.89)	0.63 (0.63, 0.64)
Logistic regression	0.70 (0.7, 0.7)	0.64 (0.63, 0.64)	0.83 (0.83, 0.83)	0.64 (0.63, 0.64)	0.63 (0.62, 0.63)
SVM linear kernel	0.7 (0.7, 0.7)	0.63 (0.63, 0.63)	0.83 (0.83, 0.83)	0.63 (0.63, 0.63)	0.49 (0.46, 0.52)
SVM RBF kernel	0.74 (0.74, 0.74)	0.69 (0.69, 0.69)	0.82 (0.82, 0.82)	0.69 (0.69, 0.69)	0.48 (0.46, 0.49)
KNN	0.69 (0.69, 0.69)	0.63 (0.62, 0.63)	0.81 (0.81, 0.81)	0.63 (0.62, 0.63)	0.55 (0.55, 0.55)
Naïve Bayes	0.03 (0.03, 0.03)	0.12 (0.12, 0.12)	0.82 (0.77, 0.86)	0.12 (0.12, 0.12)	0.50 (0.5, 0.5)
AdaBoost	0.84 (0.83, 0.84)	0.86 (0.85, 0.86)	0.82 (0.82, 0.82)	0.86 (0.85, 0.86)	0.62 (0.61, 0.63)
LSTM	0.79 (0.77, 0.81)	0.77 (0.74, 0.8)	0.82 (0.82, 0.82)	0.77 (0.74, 0.8)	0.61 (0.61, 0.61)

ML, machine learning; CI, confidence interval; AUROC, area under the receiver operating characteristic curve; MLP, multilayer perceptron; XGBoost, extreme gradient boosting; RF, random forest; SVM, support vector machine; RBF, radial basis function; KNN, k-nearest neighbor; AdaBoost, adaptive boosting; LSTM, long short-term memory.

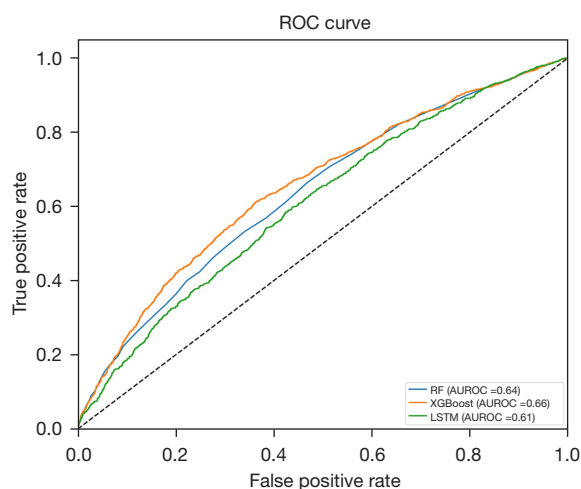


Figure 2 ROC curves for RF, XGBoost, and LSTM. ROC, receiver operating characteristic; RF, random forest; AUROC, area under the receiver operating characteristic curve; XGBoost, extreme gradient boosting; LSTM, long short-term memory.

CI: 0.82, 0.83) and 0.83 for XGBoost (95% CI: 0.83, 0.84). High precision values were observed for XGBoost classifier and RF, suggesting their effectiveness in minimizing false positives. The RF model, with the highest precision, stands out as particularly adept at correctly classifying positive instances.

Recall score, assessing a model's capacity to capture

all positive instances, exhibits similar trends to precision. XGBoost classifier and RF achieved high recall scores, indicating their ability to correctly identify a significant portion of actual positive cases. Naïve Bayes, on the other hand, displays lower recall, suggesting a higher likelihood of missing positive instances in its predictions.

AUROC, a measure of a model's performance across various threshold settings, demonstrated similar trends to recall score and the other metrics. RF achieved high AUROC (0.63, 95% CI: 0.63, 0.64), indicating its superior ability to distinguish between positive and negative classes.

In summary, by comparing the F1 score, accuracy, precision score, recall score, and AUROC, the RF model emerged as the most favorable choice for predicting diabetes readmissions in this context. It consistently outperformed other models across multiple performance metrics, showcasing a robust balance between precision and recall, ultimately contributing to the most accurate and reliable predictions.

ROC curves were plotted for RF, XGBoost, and LSTM models in *Figure 2*. The proximity of the ROC for RF to the upper left corner indicated its ability to correctly identify positive and negative cases correctly, indicating high discriminative ability. RF and XGBoost had AUROCs of 0.63 (95% CI: 0.63, 0.64) and 0.64 (95% CI: 0.64, 0.65) respectively, superior to the LSTM model (0.61, 95% CI: 0.61, 0.61).

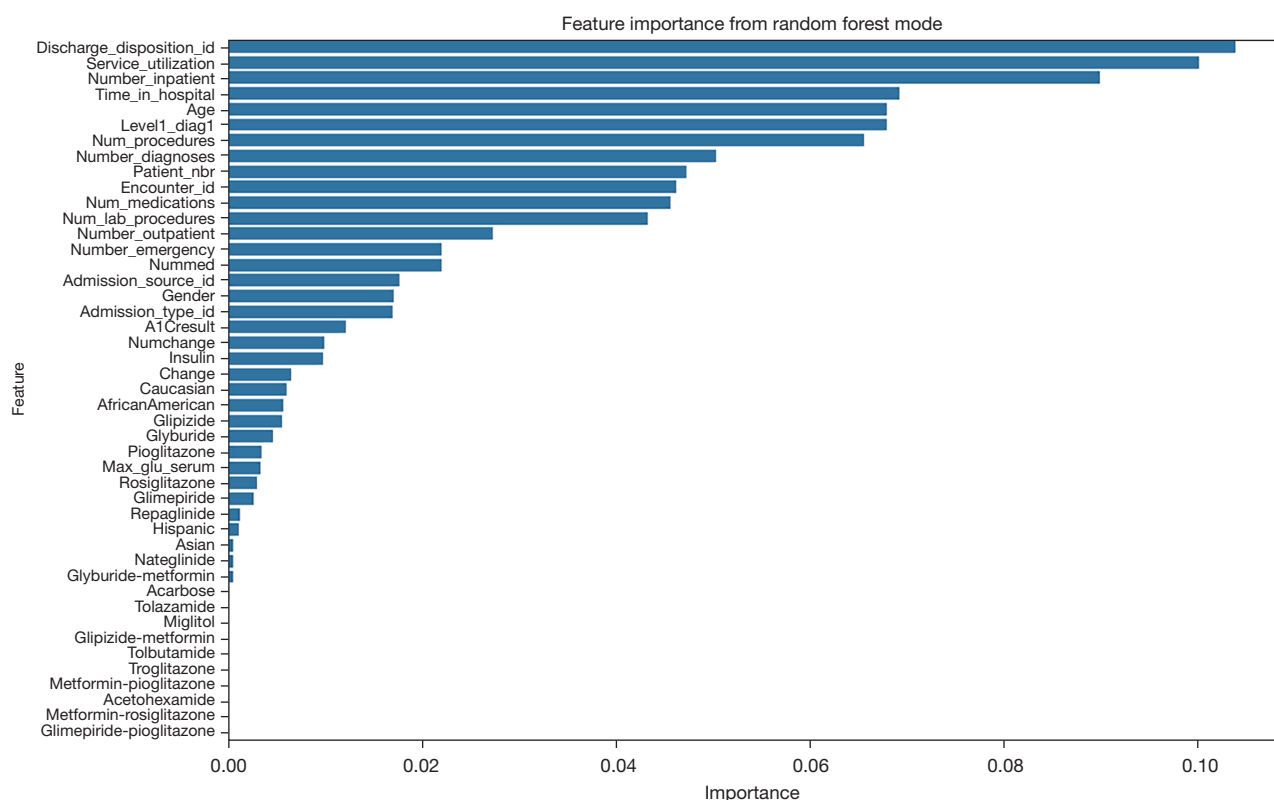


Figure 3 Graphical representation of features ranked by importance in RF model using Gini gain. RF, random forest.

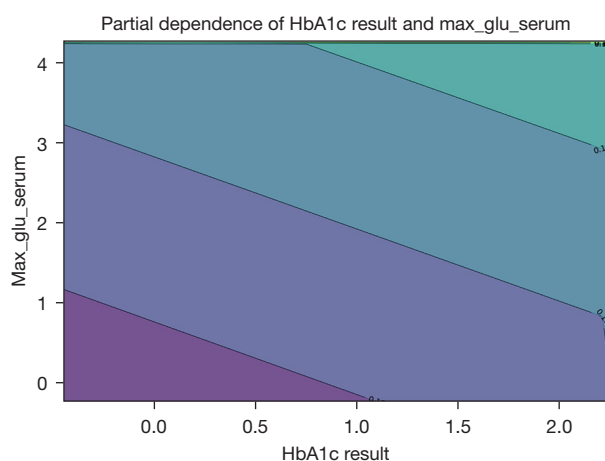


Figure 4 Partial dependence plot of HbA1c and serum glucose on readmission rate. Darker, purple shades represent lower readmission and lighter, bluer shades represent higher. HbA1c, hemoglobin A1c.

We found that the type of discharge was most important in predicting readmission status based on feature importance from the RF model using Gini gain. Demographic factors were also important, in the order of age, gender, and ethnicity. Details of the initial hospitalization and other clinical factors were also important, including time spent in hospital, number of procedures, number of medications, number of lab procedures, and the created service utilization variables. HbA1c test results, insulin, and metformin were important in that order. Other lab results ranging from glipizide to metformin-rosiglitazone were of lesser importance in predicting readmissions (*Figure 3*).

Figure 4 shows the partial dependence visualization of HbA1c and serum glucose on readmission rate. HbA1c measurements of '>7%' and '>8%' were encoded as 1, while regular levels were encoded as 0. Also, serum glucose measurements of more than 200 and 300 mg/dL were

encoded as 1, with normal blood glucose measurements encoded as 0. In the visualization, purple shades represent lower readmission rates, while blue shades represent higher readmission rates. Thus, higher levels of both blood glucose and HbA1c were associated with a higher likelihood of readmission.

Discussion

Key findings

The landscape of predicting hospital readmission risk among diabetes patients has evolved with the advent of advanced statistical modeling methods of ML, particularly its advanced DL algorithms. In the pursuit of an optimal modeling approach, several published studies have explored and compared different methodologies to enhance the accuracy and efficiency of predicting readmission risks. Comparing model performance of the 10 selected GWO-optimized ML algorithms, the results of this study indicate that RF clearly stood out as the best algorithm in multiple performance metrics, followed by the XGBoost classifier.

Strengths and limitations

A strength of this study is that it compared many different ML algorithms to predict diabetes readmission rates. Furthermore, it utilized a large dataset with a diverse patient population and significant sample size. A limitation of the study is that the dataset represents one dataset from the UCI Machine Learning Repository for 130 hospitals and integrated delivery networks from years 1999 to 2008. The Hispanic representation is low, and the length of stay in the hospital was only between 1 and 14 days. Although the electronic health records dataset comprises hospitals from each of the four geographic regions (Midwest, Northeast, South, and West) of the United States, the sample may still not be representative of more recent patients with diabetes. Although statistically significant, predictors such as blood serum glucose can vary from day to day for patients. Another limitation is that for the study to be useful clinically, there must be integration of these models in real-time hospital settings to predict patients most likely for readmission.

Comparison with similar research

Our findings of RF being superior are consistent with that

of a few studies in the literature: Thenappan *et al.* [2023] utilized ML classifiers to reduce the risk of readmission for diabetes patients (27). The study employed logistic regression, RF, and SVMs on a dataset containing patient demographic information, medical history, and laboratory results. The RF model emerged as the most precise, achieving an 83% precision rate. The study emphasized the significance of personalized diabetes management considering individual patient characteristics and risk factors. Also, a study by Shang *et al.* [2021] analyzed over 100,000 records of patients with diabetes, comparing ML classifiers of RF, Naïve Bayes, and decision tree ensemble (8). The RF model also exhibited the best performance, suggesting its suitability for predicting 30-day readmissions in patients with diabetes.

Our algorithms performed better than that from Rajput and Alashetty [2022], who proposed a ML approach with a novel preprocessing technique to reduce the readmission risk for diabetes patients (28). By using ML algorithms like KNN, SVMs, decision trees, RF, logistic regression, gradient boosting, and Gaussian Naïve Bayes, the study achieved 77.4% accuracy, compared with 88% from RF and from XGBoost classifier in our study. Also, Cui *et al.* [2018] presented an improved SVM-based method for predicting readmissions in patients with diabetes (13). Their method achieved an accuracy of 81.02% (sensitivity of 82.89%, and specificity of 79.23%), outperforming other popular algorithms, but was slightly inferior to GWO-optimized RF and XGBoost classifier in our study.

Wang and Zhu [2022] conducted a nationwide study using hospital admission data statistics to predict disease-specific 30-day readmissions (29). They employed an ensemble learning framework and identified variations in readmission rates among different diseases, emphasizing the importance of understanding disease-specific factors in predicting readmissions, which is further confirmed by our findings with serum glucose, HbA1c, and other disease measures in predicting hospital re-admission.

Our findings of XGBoost as a competitive alternative is also consistent with that from Cuong and Wang [2021] who compared various machine and DL methods, including logistic regression, artificial neural network, Naïve Bayesian classifier, SVM, and XGBoost (30). The XGBoost model, implemented in Amazon SageMaker, demonstrated the best performance, underscoring its efficacy in predicting hospital readmissions for patients with diabetes.

Our findings of the relationships between readmission and patient demographics are also consistent with those

from Alloghani *et al.* [2019] who applied ML to recognize patterns and combinations of factors associated with readmission among diabetes patients (31). The study used classifiers like linear discriminant analysis, RF, KNN, Naïve Bayes, J48, and SVM. Their findings indicated that patients more likely to be readmitted were women, Caucasians, outpatients, and those undergoing less rigorous lab procedures. However, their findings of receiving less medication being associated with higher re-admission contrast with our findings.

Notably, our findings contrasted with those of Hai *et al.*, who found that DL models such as LSTM had higher AUROC (0.79) than traditional ML models such as RF (0.72), $P < 0.0001$, while we found that traditional ML methods remained strong. This discrepancy could be due to Hai *et al.* (4) using a dataset from the Temple University Health System hospitals, encompassing a staggering number of 2,836,569 encounters, which allowed the DL model to perform better.

Explanations of findings

Effective follow-up treatment can assist to reduce readmissions and related medical expenses (32). Efforts to minimize potentially unnecessary hospitalizations for diabetes should thus target not just the overall population with diabetes, but also, and more particularly, individuals who have previously had at least one hospital stay.

In short, the studies in current literature collectively highlight the growing role of advanced ML and DL algorithms in predicting hospital readmission risks among diabetes patients. While various models demonstrated effectiveness, the DL LSTM model by Hai *et al.* [2022] and our GWO-optimized algorithms stood out, showcasing significantly superior performance compared to traditional models, making them notable candidates for consideration in future predictive models (4).

Our new study introduced model comparison with key metrics such as F1, AUROC, accuracy, precision score, recall score, and AUROC. In the XGBoost classifier, the model achieves a high F1 score of 0.84, and an accuracy of 0.88. The precision and recall scores are also noteworthy at 0.84 and 0.88, respectively.

Although the best models were obtained, they should be further refined to include additional clinical parameters that may enhance prediction of meaningful outcomes. For example, it has been well-documented that hypoglycemia and hyperglycemia are associated with

adverse patient outcomes. A study by Zapatero *et al.* found that hypoglycemia in diabetes patients in Spanish internal medicine wards was associated with early readmission, in addition to increased length of stay and inpatient mortality (33). Cichosz *et al.* found that spontaneous hypoglycemia in intensive care units (ICUs) led to an increased mortality rate of 15.6% in those patients *vs.* 8% in patients without hypoglycemia. With respect to hyperglycemia, it was strongly associated with adverse events in patients without diabetes in ICUs, while the impact was smaller in patients with diabetes (34). A possible explanation provided for this discrepancy is that surveillance of glucose is more prevalent in patients with diabetes, which may have led to increased identification of hyperglycemia (and therefore potentially earlier intervention or treatment, reducing adverse outcomes) (34). Strack *et al.* also explained that monitoring HbA1c appeared to be associated with lower readmission rates when the primary diagnosis was diabetes because of increased level of care and attention paid to diabetes when it was the primary diagnosis (9). A summary of other studies detailing the impact of HbA1c on 30-day readmission can be found in Rubin (2). Future work using the best ML models for examining the effects of hypoglycemia and hyperglycemia, via HbA1c measure, point of care glucose testing, and/or continuous glucose monitoring, is warranted in patients both with and without diabetes.

Implications and actions needed

Comparing these results with the metrics presented previous works outlined above, it is evident that the new model, represented by RF and XGBoost classifier, demonstrated competitive performance. The AUROC, accuracy, and precision score values were notably high, indicating robust predictive capabilities. In contrast to previous studies, RF outperformed several traditional models cited in the literature.

Across various studies, the presented models have utilized different algorithms such as DL, RF, SVMs, and gradient boosting, each tailored to specific datasets and objectives. However, the new model, based on the RF and XGBoost classifier, appears to achieve comparable or superior results in terms of accuracy and predictive power.

While it is crucial to acknowledge the diversity in datasets, methodologies, and goals across these studies, the RF and XGBoost models set a high standard in terms of its overall performance metrics. Whether this sets a new

standard or target for the field will depend on the specific context and goals of the predictive models.

Advancements in ML models for predicting hospital readmissions, as reflected in the RF and XGBoost results, suggest progress in the field. The high accuracy and precision scores indicate improved capabilities in identifying patients at risk of readmission. However, the decision for hospitals to adopt these new models should be carefully considered in the context of their specific patient populations, resources, and the overall goals of readmission prevention. Identifying other important clinical or demographic risk factors for hospital readmission, and including these in future modeling studies, is also important, with an ultimate goal of reducing hospital readmission for patients with diabetes.

In conclusion, the new RF and XGBoost model present promising results with high accuracy and precision. While they may not establish an entirely new standard, it certainly raises the bar for predictive modeling in hospital readmission scenarios. Hospitals should adjust the positive class threshold for readmissions based on their specific needs, balancing more false positives against fewer false negatives to find the optimal threshold. Higher hospital readmission rates are associated with an increased disease burden for patients with diabetes and have important medical, social, quality of life, and financial implications (5), so predictive modeling could be a useful tool in predicting which patients are at higher risk for readmission and designing programs to reduce this risk. These could include close outpatient follow-up following hospital discharge (with the patient's primary care provider or endocrinologist), home health visits, telemedicine follow-up shortly after discharge, or other interventions (3). Hospitals should assess the compatibility of the model with their specific needs before considering adoption, recognizing the ongoing advancements in the field of predictive analytics for healthcare.

Conclusions

The vast amount of electronic health records data available can be used to create robust models for predicting hospital readmission rates. Our work with one such dataset, from the UCI ML repository, of patients from 130 hospitals across the United States, showed that the RF and XGBoost models, when enhanced with GWO for feature selection, group k-fold cross-validation and SMOTE, and feature engineering, produced models with F1 scores rivaling or

outperforming those models presented in the literature. Our work also demonstrated the hierarchy of the eleven examined ML models from best to worst for the prediction. Future work would involve applying these trained models to patients with diabetes in a real-time hospital setting, identifying those that are most at-risk for being readmitted based on a combination of their demographics, admission status, and laboratory tests. These data-driven approaches to improving healthcare show high promise and in fact already demonstrate remarkable accuracy and utility. Thus, hospitals can be encouraged to begin using these models to improve patient care and ultimately outcomes for patients with diabetes to reduce diabetes related co-morbidities and complications.

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Footnote

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Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at <https://jmai.amegroups.com/article/view/10.21037/jmai-24-70/coif>). The authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. IRB and informed consent were not required as this study does not involve any human experiments.

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